# Exercises: Duality Theory and KKT Conditions

Mathematical Foundations for Machine Learning

## Lagrangian Formulation and KKT Conditions

**Exercise 1.1** [Easy]
Consider the optimization problem

$$\min_{x \in \mathbb{R}^2} x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \geq 2.$$

complementary slackness is g(x)=0so if g(x) activ

**(a)** Rewrite the constraint in the standard form $g(x) \leq 0$.
**(b)** Write down the Lagrangian $\mathcal{L}(x, \lambda)$.
**(c)** State all four KKT conditions for this problem explicitly (do not solve them yet).

**Exercise 1.2** [Easy]
Consider the problem from Exercise 1.1. A candidate solution is $x^* = (1, 1)$ with multiplier $\lambda^* = 2$.

in convex local min -> global min

**(a)** Verify that all four KKT conditions are satisfied at $(x^*, \lambda^*)$. if we find point that sat. all -> optim. point

**(b)** Is the objective function convex? Is the feasible set convex? What can you conclude about $x^*$?

**(c)** Give an interpretation of the value of $\lambda^* = 2$: if the constraint is changed to $x_1 + x_2 \geq 2 + \epsilon$ for small $\epsilon > 0$, by approximately how much does the optimal objective value change?

The bigger

**Exercise 1.3** [Medium]
Formulate an optimization problem and use the KKT conditions to find the point in the set

$$M = \{x \in \mathbb{R}^2 : x_1^2 + x_2 \leq 4,\ x_1 - x_2 \leq 0\}$$

closest to the point $p = (0, 5)$, using the squared Euclidean distance as objective.

**(a)** Write the optimization problem in standard form.
**(b)** Write the Lagrangian and all KKT conditions.
**(c)** Use complementary slackness to determine which constraints are active, and solve for the optimal point.
**(d)** Verify that your solution is indeed a global minimizer.

**Exercise 1.4** [Medium]
Consider the problem of minimizing the squared norm on the probability simplex:

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^{n} x_i^2 \quad \text{s.t.} \quad \sum_{i=1}^{n} x_i = 1, \quad x_i \geq 0\ \forall i.$$

**(a)** Write the Lagrangian with multiplier $\mu$ for the equality constraint and $\lambda_i \geq 0$ for the inequality constraints $-x_i \leq 0$.

**(b)** Write the KKT conditions and use complementary slackness to argue that $x_i^* = \frac{1}{n}$ for all $i$.

**(c)** What is the optimal value? How does it behave as $n \to \infty$? Give an interpretation in the context of diversification in portfolio optimization.

**Exercise 1.5** [Hard]
**Ridge regression as constrained optimization.** Consider the problem

$$\min_{\theta \in \mathbb{R}^D} \|Y - X\theta\|_2^2 \quad \text{s.t.} \quad \|\theta\|_2^2 \leq t,$$

In the dual problem, instead of minimizing accounting for all the constraints, we are maximizing from below. We first compute all the possible minimizers for all constraints, and then get the constraint that gets the maximum minimizer. So we take that and we can say that it is smaller than the real minimizer of the feasible region (equal if strong duality).

If there is a feasible point that holds all the inequalities in a strict manner then the lagragian penalty mechanism can smoothly approximate it -> strong duality holds.



where $X \in \mathbb{R}^{N\times D}$, $Y \in \mathbb{R}^N$, and $t > 0$.

**(a)** Write the Lagrangian and derive the KKT conditions.

**(b)** From the stationarity condition, show that the optimal $\theta^*$ satisfies

$$\theta^* = (X^T X + \lambda I)^{-1} X^T Y$$

for some $\lambda \geq 0$, and explain why $X^T X + \lambda I$ is always invertible for $\lambda > 0$.

**(c)** Use complementary slackness to describe when $\lambda = 0$ (no regularization) vs. $\lambda > 0$ (active regularization). Relate this to the OLS estimator $\hat{\theta}_{\text{OLS}} = (X^T X)^{-1} X^T Y$.

**(d)** Prove that this constrained problem is convex and that strong duality holds (verify Slater's condition).

## Dual Function and Dual Problem

**Exercise 2.1** [Easy]
Consider the one-dimensional problem

$$\min_{x\in\mathbb{R}} x^2 \quad \text{s.t.} \quad x \geq 2.$$

**(a)** Rewrite in standard form and compute the Lagrangian.

**(b)** Derive the dual function $q(\lambda)$ by minimizing the Lagrangian over $x$.

**(c)** State the dual problem and solve it. What is $d^*$? What is $p^*$? Is there a duality gap?

**Exercise 2.2** [Medium]
**Dual of least-squares with linear constraints.** Consider

$$\min_{x\in\mathbb{R}^n} \|x\|_2^2 \quad \text{s.t.} \quad Ax = b,$$

where $A \in \mathbb{R}^{p\times n}$ with $p < n$ and $b \in \mathbb{R}^p$.

**(a)** Write the Lagrangian $\mathcal{L}(x, \mu)$.

**(b)** Derive the dual function $q(\mu)$ by solving the stationarity condition $\nabla_x \mathcal{L} = 0$.

**(c)** Write the dual problem explicitly. Show it is a concave quadratic in $\mu$.

**(d)** Assuming $AA^T$ is invertible, solve the dual problem and recover the primal optimal solution $x^*$.

**Exercise 2.3** [Medium]
**Dual of a linear program.** Consider the LP in inequality form:

$$\min_{x\in\mathbb{R}^n} c^T x \quad \text{s.t.} \quad Ax \leq b,$$

where $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$.

**(a)** Write the Lagrangian $\mathcal{L}(x, \lambda)$ with $\lambda \in \mathbb{R}^m$, $\lambda \geq 0$.

**(b)** Derive the dual function $q(\lambda) = \inf_x \mathcal{L}(x, \lambda)$. Identify the hidden constraint that must hold for $q(\lambda) > -\infty$.

**(c)** Write the dual problem explicitly, making the hidden constraint explicit. Show that the dual of a linear program is again a linear program.

**Exercise 2.4** [Hard]
**Dual of the Support Vector Machine.** Consider the hard-margin SVM problem with training data $\{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathbb{R}^D$ and $y_i \in \{-1, +1\}$:

$$\min_{w\in\mathbb{R}^D} \frac{1}{2}\|w\|_2^2 \quad \text{s.t.} \quad y_i(w^T x_i) \geq 1, \quad i = 1, \ldots, N.$$

**(a)** Rewrite the constraints in the form $g_i(w) \leq 0$ and write the Lagrangian.

**(b)** Derive the dual function by minimizing the Lagrangian over $w$. Express it purely in terms of the multipliers $\alpha_i$ and the data.

**(c)** Write the dual problem. Show that the dual depends on the data only through the inner products $\langle x_i, x_j \rangle$. Why is this important?

**(d)** State the complementary slackness conditions and explain why data points with $\alpha_i > 0$ are called *support vectors*.

**(e)** This problem is convex. Argue why Slater's condition holds if the data is linearly separable, and conclude that strong duality holds.

## Weak Duality, Strong Duality, and Slater's Condition

**Exercise 3.1 [Easy]**
Consider the problem

$$\min_{x \in \mathbb{R}} (x-3)^2 \quad \text{s.t.} \quad x \geq 1.$$

**(a)** What is the primal optimal value $p^*$?

**(b)** Compute the dual function $q(\lambda)$ and evaluate it at $\lambda = 1$. Verify that $q(1) \leq p^*$ (weak duality).

**(c)** Prove the weak duality theorem in general: for any $\lambda \geq 0$, $q(\lambda) \leq p^*$.

**Exercise 3.2 [Medium]**
Consider the problem

$$\min_{x \in \mathbb{R}^2} e^{-x_1} \quad \text{s.t.} \quad \frac{x_1^2}{x_2} \leq 0, \quad x_2 > 0.$$

We restrict to the domain $\mathcal{D} = \{x \in \mathbb{R}^2 : x_2 > 0\}$.

**(a)** Show that the primal optimal value is $p^* = 1$.

**(b)** Compute the dual function $q(\lambda)$ and the dual optimal value $d^*$.

**(c)** Is there a duality gap? Does Slater's condition hold? Explain the connection.

**Exercise 3.3 [Medium]**
Consider the convex optimization problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad Ax = b, \quad g_i(x) \leq 0, \ i = 1, \ldots, r,$$

where $f$ and all $g_i$ are convex. Suppose Slater's condition holds (there exists $\tilde{x}$ with $A\tilde{x} = b$ and $g_i(\tilde{x}) < 0$ for all $i$).

**(a)** State the consequence of Slater's condition for this problem.

**(b)** Prove that for a convex problem with strong duality, if the primal optimum $x^*$ and dual optimum $(\lambda^*, \mu^*)$ are attained, then the KKT conditions hold at $(x^*, \lambda^*, \mu^*)$.

(Hint: Use $f(x^*) = p^* = d^* = q(\lambda^*, \mu^*) = \inf_x \mathcal{L}(x, \lambda^*, \mu^*)$ and argue about the chain of inequalities.)

**Exercise 3.4 [Hard]**
**When strong duality fails.** Consider the non-convex problem

$$\min_{x \in \mathbb{R}} x \quad \text{s.t.} \quad x^2 \leq 0.$$

**(a)** Find the primal optimal value $p^*$.

**(b)** Compute the dual function $q(\lambda)$ and the dual optimal $d^*$.

**(c)** Show there is a duality gap and explain why. Which assumption of Slater's theorem is violated?

**(d)** Now consider the modified problem $\min_x x$ s.t. $x^2 - 1 \leq 0$. Show that Slater's condition holds, compute the dual function and dual optimal value, and verify strong duality.

## Mixed Challenging Exercises

**Exercise 4.1 [Hard]**
**Entropy maximization and KL divergence (Exam-style).** Let $p = (p_1, \ldots, p_n)$ be a fixed probability distribution with $p_i > 0$ for all $i$. Consider the problem

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^{n} x_i\big(\log(x_i) - \log(p_i)\big) \quad \text{s.t.} \quad \sum_{i=1}^{n} x_i = 1, \quad x_i \geq 0 \; \forall i.$$

Note: this objective is the KL divergence $D_{\mathrm{KL}}(x \| p)$.

**(a)** Show that $\phi(x) = \sum_{i=1}^{n} x_i(\log x_i - \log p_i)$ is strictly convex on $\{x : x_i > 0\}$.

**(b)** Write the Lagrangian with multiplier $\mu$ for the equality constraint and $\lambda_i \geq 0$ for $-x_i \leq 0$. Derive all KKT conditions.

**(c)** Using complementary slackness and stationarity, show that $x_i^* = p_i$ for all $i$ and determine the multipliers.

**(d)** What is the optimal value? Explain why this is consistent with the known property $D_{\mathrm{KL}}(q \| p) \geq 0$ with equality iff $q = p$.

**(e)** Assuming strong duality holds, derive the dual function and solve the dual problem. Verify that the dual optimal value equals the primal optimal value.

**(f)** *(Bonus)* The problem also has a saddle-point interpretation: $p^* = \inf_x \sup_{\lambda \geq 0, \mu} \mathcal{L}(x, \lambda, \mu)$. Without proof, state the corresponding expression for $d^*$ and the condition under which they are equal.

**Exercise 4.2 [Hard]**
**Constrained quadratic optimization with multiple constraints.** Consider

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^T Q x + c^T x \quad \text{s.t.} \quad Ax \leq b, \quad Cx = d,$$

where $Q \in \mathbb{R}^{n \times n}$ is symmetric positive definite, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{p \times n}$, $d \in \mathbb{R}^p$.

**(a)** Write the Lagrangian $\mathcal{L}(x, \lambda, \mu)$ with $\lambda \in \mathbb{R}_+^m$ and $\mu \in \mathbb{R}^p$.

**(b)** Derive the dual function $q(\lambda, \mu)$ in closed form.

**(c)** Write the dual problem. Explain why it is always a concave maximization problem, regardless of the original problem's convexity.

**(d)** Prove that Slater's condition holds if there exists $\tilde{x}$ with $A\tilde{x} < b$ (componentwise) and $C\tilde{x} = d$. Conclude strong duality.

**(e)** For the specific instance $n = 2$, $Q = I$, $c = 0$, with constraints $x_1 + x_2 \leq 1$ and $x_1 - x_2 = 0$: use the KKT conditions to find the optimal solution, the active constraints, and all multipliers.

**Exercise 4.3 [Hard]**
**Maximum likelihood with constraints.** Consider $N$ observations $y_1, \ldots, y_N$ drawn i.i.d. from a categorical distribution over $\{1, \ldots, K\}$ with probabilities $\theta = (\theta_1, \ldots, \theta_K)$. Let $n_k = |\{i : y_i = k\}|$ be the count of category $k$. The log-likelihood is

$$\ell(\theta) = \sum_{k=1}^{K} n_k \log \theta_k.$$

**(a)** Write the constrained MLE problem as a maximization with constraints $\sum_k \theta_k = 1$ and $\theta_k \geq 0$.

**(b)** Convert to a minimization problem. Write the Lagrangian and KKT conditions.

**(c)** Using complementary slackness, show that the MLE is $\hat{\theta}_k = n_k/N$.

**(d)** Derive the dual function and dual problem. Verify strong duality by showing the optimal dual value equals $-\ell(\hat{\theta})$.

**(e)** The dual of MLE connects to exponential families. Show that the stationarity condition can be written as $\mathbb{E}_{\hat{\theta}}[\mathbf{1}_{Y=k}] = n_k/N$, i.e., the model's expected sufficient statistics match the empirical ones. This is the *moment matching* property of MLEs in exponential families.